higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 3

(12020353)

11 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: GENERAL

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 The process during which concrete gains strength is called stress.

1.2 There is no difference between screed and topping.

1.3 Racking back is the laying of stepped courses of bricks meant to assist the bonding of masonry.

1.4 Brick force provides tension reinforcement in masonry work to control cracking.

1.5 Stretcher bond is also known as Flemish bond.

1.6 Lintels are built into a wall to support the foundation.

1.7 A damp-proof course is a barrier in a wall that prevents moisture rising up through the building.

1.8 A cavity wall consists of a single-leaf wall built with hollow concrete blocks.

1.9 The CMA stipulates that units shall be laid in stretcher bond unless otherwise stated.

1.10 A slump test is used to test the consistency of a brick.

(10 × 1) **[10]**

QUESTION 2: CAVITY WALL CONSTRUCTION

2.1 State the difference between each of the following:

2.1.1 DPC

2.1.2 DPM

(2 × 2) (4)

2.2 Name FOUR types of wall ties suitable for cavity walls. (4)

2.3 What is the purpose of weep holes in a cavity wall? (2)

2.4 Name SIX places where DPC must be placed when constructing a building. (6)

2.5 Briefly discuss FIVE ways to build a lintel over a cavity wall. (5)

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2.6 Draw, to a suitable scale, a vertical section through the bottom 10 courses of a 270 mm cavity wall.

Clearly show the following:

- Concrete strip foundation: 600 mm × 200 mm
- Hard core: 75 mm
- Floor slab: 75 mm
- Cavity wall
- Screed: 20 mm
- Internal plaster: 19 mm
- Ground level
- Bottom EIGHT courses
- Damp-proof courses (9)

[30]

QUESTION 3: BASIC MASONRY WALL CONSTRUCTION WITH HOLLOW UNITS

3.1 Name THREE materials used to manufacture hollow masonry units. (3)

3.2 Name FOUR guidelines to build straight walls using hollow units. (4)

3.3 It sometimes happens that foundations are left uneven and not level. This causes walls to not be straight and bed joints not level.

Briefly explain how to check if a foundation is level and even. (10)

3.4 Make a list of the tools used to rule each of the following mortar joints in a hollow-block masonry wall:

- Concave joint
- V-joint
- Flush joint
- Raked joint
- Weathered joint (5)

3.5 Name FOUR types of hollow masonry units and indicate where the units will be used. (4 × 2) (8)

[30]

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QUESTION 4: ADVANCED MASONRY WALL CONSTRUCTION

4.1 What is the specified depth for roof anchors for lightweight and heavyweight roofs? (2)

4.2 Briefly explain each the following arch terms:

4.2.1 Voussoirs

4.2.2 Span

4.2.3 Keystone

4.2.4 Extrados

4.2.5 Crown

4.2.6 Intrados

(6 × 2) (12)

4.3 Name TWO types of roof anchors. (2)

4.4 According to SANS 2001:CM1, how long should a lintel be propped before the props can be removed? (2)

4.5 Describe the implications if a lintel is built without detailed drawings. (2)

4.6 Explain how to clean clay masonry without scratching the bricks. (5)

4.7 Name FIVE tools needed to construct an arch in face brick. (5)

[30]

TOTAL: 100

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